

Supplement for RDD results

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Abstract

Supplemental material to support the RDD result in Arjmandi-Lari, Z., Mantzarlis, A., Stafford, T. (2025). [Threats to the sustainability of Community Notes on X](#). arXiv preprint arXiv:2510.00650

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1 Introduction

Find all of this in the repo https://github.com/zahra-arjm/community_notes

This PDF made by a quarto document <supplement.qmd>. The quarto file renders the outputs for the full set of RDD results (sensitivity check etc), and also pushes some outputs to /paper-outputs as a .tex fragments so they can be pulled into overleaf directly.

To see the code which is running you need to open the .qmd file. As a PDF this shows commentary (like this text) and code outputs (in text, and images).

AI declaration: all the words are human-written, all the logic and claims are human endorsed (by TS, so they may contain errors). The code was co-written with using an AI agent, some comments from the agent are left in the code (invisible in the PDF).

1.1 Topline results and v1-v2 chanegs

- Since v1 we adopt a more sophisticated RDD method, fix bugs and are stricter on best practice with respect to causal inference. There is still a positive effect, but it is not significant (confidence bounds overlap an effect of 0).
- A zero-indexing bug meant previously we were looking at the effect of first note status on 3rd note publication. This is now fixed.
- There is (remains) a robust effect for authoring a 3rd note (`have_another_note = note_counts[(note_counts >= 3)]`).
- We discover the RDD is not sharp, due to the fact that we have the score (intercept) from the data cut off point, which is NOT the intercept at the point when the note was or was not published (scores continuing to evolve post publication). This non-sharpness can be dealt with by running a fuzzy-RDD which accounts for the change in probabilities of being published at different intercept scores (there is still a discontinuity in these probailities).
- In addition there is a special population of Notes just below the cutoff threshold (intercept 0.39 - 0.40) which have a high probability of having been published. This is due to a documented *feature* of the CN system, whereby notes are protected from being unpublished when their score drops by a small amount (<0.01). These can be handled by using a “Donut” RDD, where notes +/- 0.01 of the cutoff are excluded
- To do all of this we move from using OLS to the robust RDD package
- We also run additional sensitivity checks and robustness checks (e.g. local density check around threshold)

2 Setup & Data Import

We fix the cumsum=>2 issue, so we have authors who write 2 (not 3) notes as our outcome variable
We also remove the alignment filter, since this is conditioning on a collider (and makes our RDD artificially sharp)

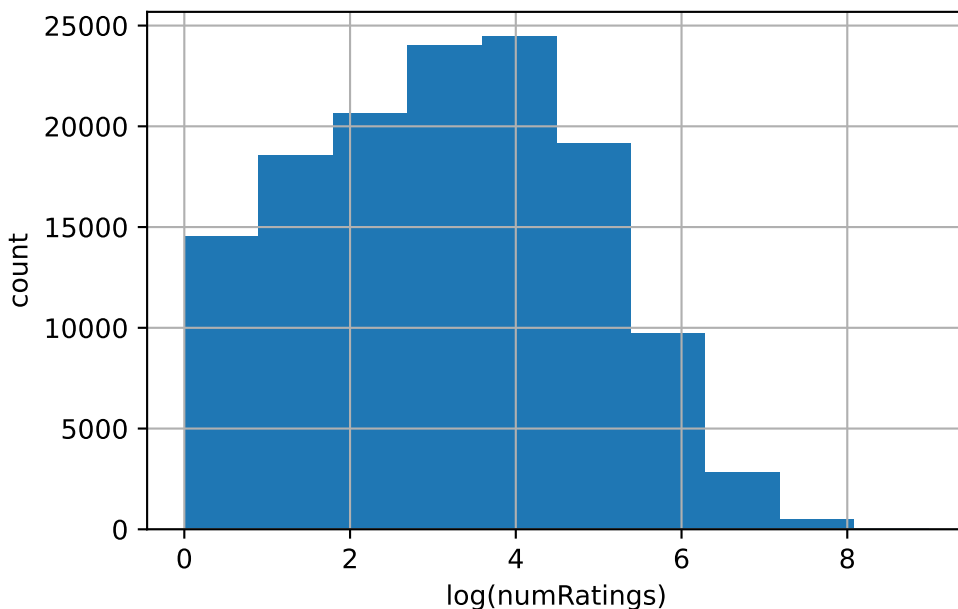
```
Imported data for 3173276 notes
ALERT! re_auth_thresh = 2
Now we have 330234 first time note authors (all time)
Now we have 226000 first time note authors 2024-01-01 to 2026-01-01
Now we have 202507 notes written by our 4 choosen algorithms
Now we have 149879 notes classified as MISLEADING
Now we have 134525 notes which have >0 ratings
----- No filter for alignment
FINAL total 134525 Notes
of which author writes again = 96269
```

NB current (final), not Max, score

We are now also using Notes evaluated by `{python} f"{len(first_notes['decidedBy'].unique())}"` models

3 Diagnostic Plots

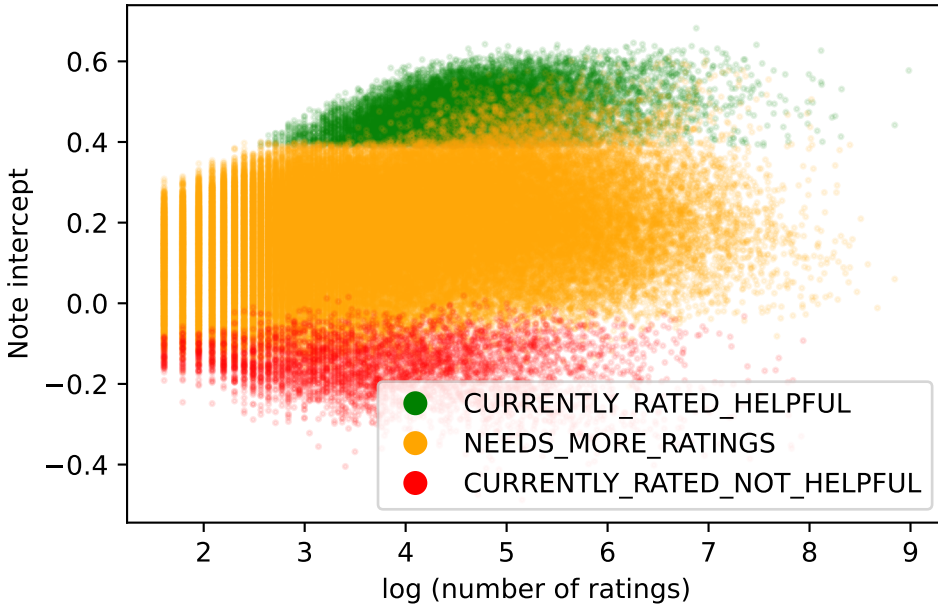
Most notes have a small number of rating, a small number have very many ratings. This means the distribution of ratings is very skewed. We take the log of the number of ratings to make the distribution more normal.



The basis for our RDD is the discontinuity between the intercept and note status (which defines publication / the possibility for publication)

Plotting $\log(\text{numRatings})$ vs NoteIntercept, with colours according to Note final status (finalRatingStatus)

You can see the fuzzy discontinuity - some $>\text{cutoff}$ notes are “NEEDS_MORE_RATINGS”



Sadly, though, our data record intercept calculated from the full set of ratings, this is not the contemporaneous intercept score which determines moment-by-moment publication. It approximates it, but with contamination. Notes may be published then unpublished as they receive more ratings and their rating degrades, or there may be additional blocks on publication (e.g. requirements on total number or type of ratings, or requirements on intercept stability or a lag before publication). The CN system is not entirely clear about all these details.

These contamination can be seen by plotting indicators of ever having been published against our (final) intercept.

We construct an independent ‘ever_published’ variable from the `firstNonNMRStatus` and `mostRecentNonNMRStatus` variables. The logic being that if either of these are `CURRENTLY_RATED_HELPFUL` then the Note must have achieved CRH/published status at some point in the past (ie separate from the `finalRatingStatus` score)

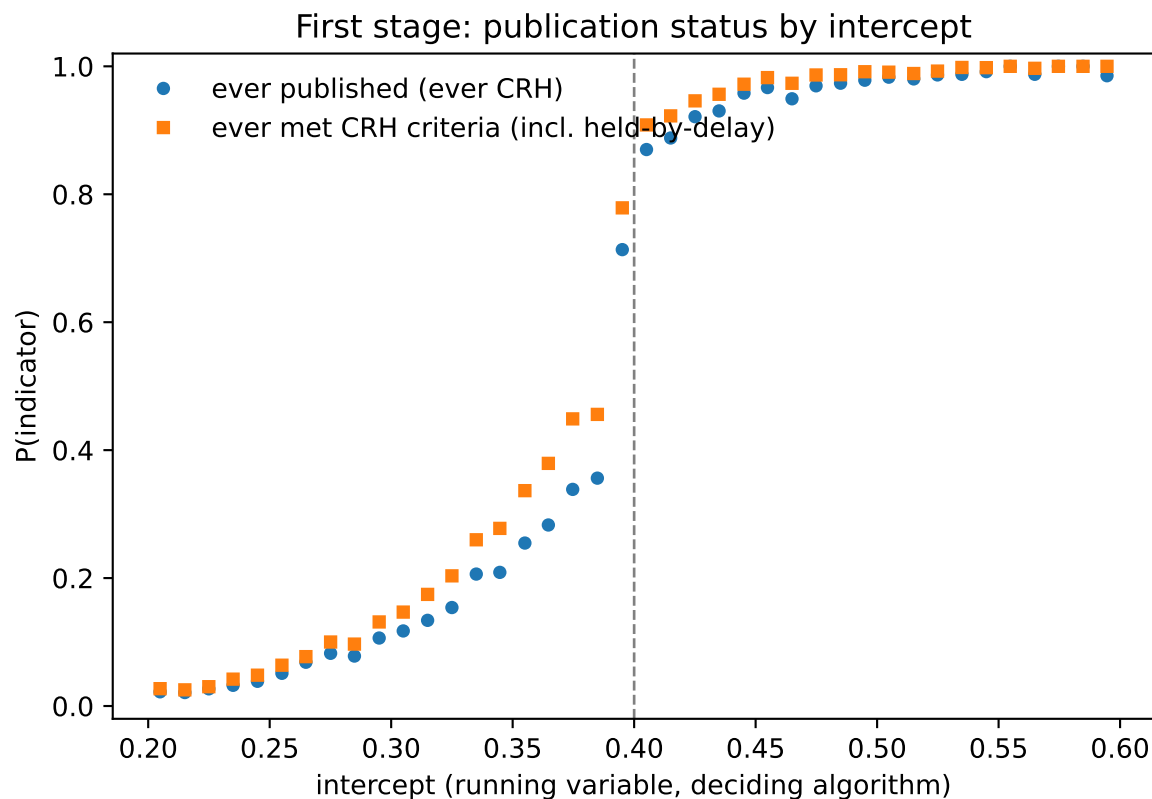
```
[0.35,0.40): n= 4233 P(ever_pub)=0.4011 P(ever_crit)=0.4895
[0.38,0.40): n= 1638 P(ever_pub)=0.5800 P(ever_crit)=0.6581
[0.39,0.40): n= 1026 P(ever_pub)=0.7135 P(ever_crit)=0.7788
[0.40,0.42): n= 1723 P(ever_pub)=0.8787 P(ever_crit)=0.9153
```

Below-cutoff `P(if_written_again)` by `ever_published`:

	mean	size
<code>ever_published</code>		
0	0.703219	89379
1	0.759502	4341

Below-cutoff `P(if_written_again)` by `ever_met_crh_criteria`:

	mean	size
<code>ever_met_crh_criteria</code>		
False	0.703886	88368
True	0.737855	5352



This shows that our secondary indicator of publication status shows a discontinuity near the cutoff, but not a sharp one

The high proportion of 'ever published' just below the cutoff suggests many notes got published and then slipped back below the threshold

We deal with this by a) using a fuzzy RDD and b) excluding the just below cutoff points by using the donut method

4 Trad OLS RDD

Our first approach was to use OLS and an arbitrary window

This is now significant with a wide window (0.08, it was non-significant at 0.05), and a 0.01 donut (but not significant if donut = 0, because of the contamination in the 0.39-0.40 zone))

bandwidth = 0.08

Number of treated units: 6336

Number of control units: 7164

Estimated Treatment Effect at the Cutoff: 0.0589

I think we can intepret this as a % change the chance of authoring again if published

P-value for treatment: 0.0041

T-statistic for treatment: 2.8721

This OLS approach provides a bridge to the fuzzy-RDD approach, which uses robust RDD to estimate the window size (as well as bringing in a bunch of statistical machinery which is best practice for RDDs)

5 RDD using fuzzy rdrobust with a donut

The rdrobust function will estimate the bandwidth from the data, as well as provide various sensitivity checks and allow a fuzzy RDD

It has the option of using a polynomial fit (which we decline), and uses all points, weighted by proximity to cutoff

We are using a ‘donut’ method of excluding those points near the cutoff, this is specifically to get those points just below the cutoff which have a high (outlier) percentage change of ever having been published

References for this method

@article{calonico2014robust, title = {Robust Nonparametric Confidence Intervals for Regression-Discontinuity Designs}, author = {Calonico, Sebastian and Cattaneo, Matias D. and Titiunik, Rocio}, journal = {Econometrica}, volume = {82}, number = {6}, pages = {2295–2326}, year = {2014}, doi = {10.3982/ECTA11757} }

@article{calonico2018effect, title = {On the Effect of Bias Estimation on Coverage Accuracy in Nonparametric Inference}, author = {Calonico, Sebastian and Cattaneo, Matias D. and Farrell, Max H.}, journal = {Journal of the American Statistical Association}, volume = {113}, number = {522}, pages = {767–779}, year = {2018}, doi = {10.1080/01621459.2017.1285776} }

@article{calonico2014rdrobust, title = {Robust Data-Driven Inference in the Regression-Discontinuity Design}, author = {Calonico, Sebastian and Cattaneo, Matias D. and Titiunik, Rocio}, journal = {The Stata Journal}, volume = {14}, number = {4}, pages = {909–946}, year = {2014}, doi = {10.1177/1536867X1401400413} }

Justifying the donut RDD, we can test if intercept (our running variable) is associated with ‘ever published’

PRIMARY SPEC - donut ± 0.01

Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations:	105375	
Polynomial Order Est. (p):	1	
Polynomial Order Bias (q):	2	
Kernel:	Triangular	
Bandwidth Selection:	mserd	
Var-Cov Estimator:	NN	
	Left	Right

Number of Observations	92694	12681

Number of Unique Obs.	92529	12672
Number of Effective Obs.	3247	3640
Bandwidth Estimation	0.05	0.05
Bandwidth Bias	0.123	0.123
rho (h/b)	0.411	0.411

Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	0.457	0.033	13.725	7.163e-43	[0.392, 0.523]
Bias-Corrected	0.441	0.033	13.246	4.780e-40	[0.376, 0.507]
Robust	0.441	0.037	12.0	3.553e-33	[0.369, 0.513]

- - - - variation: donut ±0.0

Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations:	107285
Polynomial Order Est. (p):	1
Polynomial Order Bias (q):	2
Kernel:	Triangular
Bandwidth Selection:	mserd
Var-Cov Estimator:	NN

	Left	Right
Number of Observations	93720	13565
Number of Unique Obs.	93554	13555
Number of Effective Obs.	1778	1921
Bandwidth Estimation	0.022	0.022
Bandwidth Bias	0.066	0.066
rho (h/b)	0.339	0.339

Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	0.026	0.027	0.968	3.332e-01	[-0.027, 0.079]
Bias-Corrected	0.012	0.027	0.454	6.496e-01	[-0.04, 0.065]
Robust	0.012	0.028	0.438	6.616e-01	[-0.042, 0.067]

- - - - variation: donut ±0.005

Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations:	106316
Polynomial Order Est. (p):	1
Polynomial Order Bias (q):	2
Kernel:	Triangular

Bandwidth Selection: mserd
 Var-Cov Estimator: NN

	Left	Right
Number of Observations	93176	13140
Number of Unique Obs.	93010	13131
Number of Effective Obs.	1345	1630
Bandwidth Estimation	0.024	0.024
Bandwidth Bias	0.077	0.077
rho (h/b)	0.308	0.308

Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	-0.008	0.047	-0.18	8.573e-01	[-0.1, 0.083]
Bias-Corrected	-0.033	0.047	-0.706	4.803e-01	[-0.125, 0.059]
Robust	-0.033	0.049	-0.676	4.991e-01	[-0.129, 0.063]

- - - variation: donut ± 0.02

Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations: 103924
 Polynomial Order Est. (p): 1
 Polynomial Order Bias (q): 2
 Kernel: Triangular
 Bandwidth Selection: mserd
 Var-Cov Estimator: NN

	Left	Right
Number of Observations	92082	11842
Number of Unique Obs.	91917	11833
Number of Effective Obs.	4092	3992
Bandwidth Estimation	0.063	0.063
Bandwidth Bias	0.179	0.179
rho (h/b)	0.352	0.352

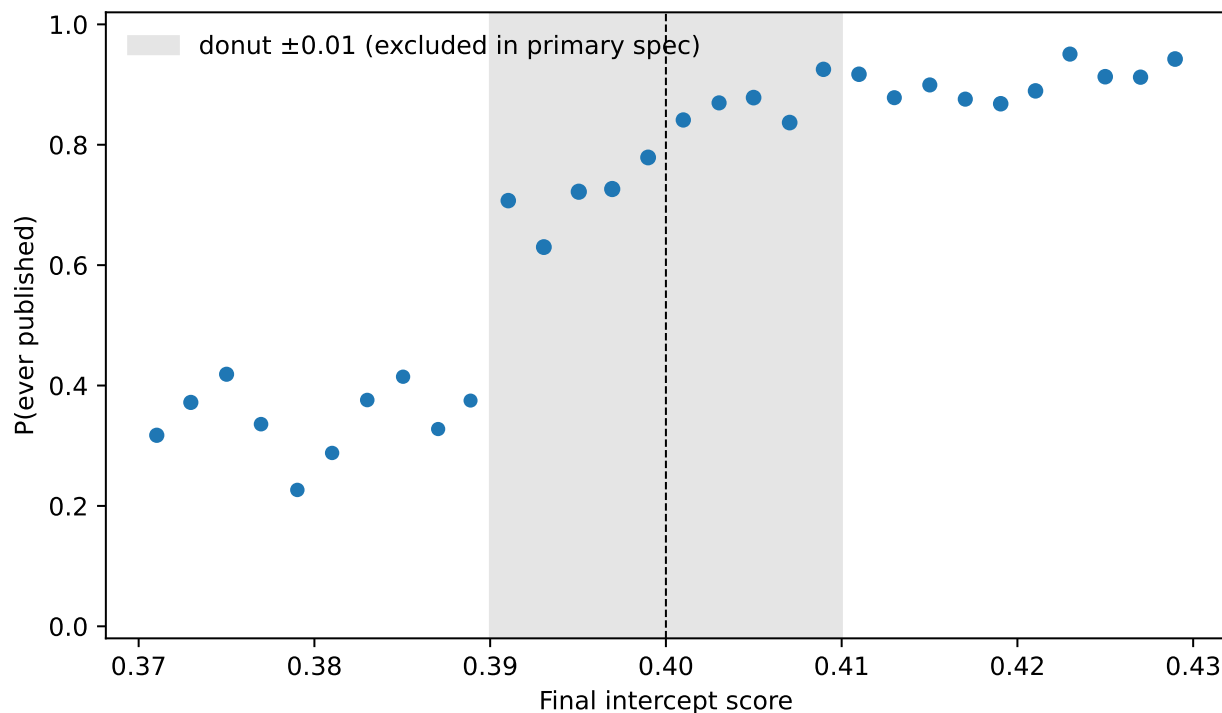
Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	0.447	0.035	12.626	1.527e-36	[0.378, 0.517]
Bias-Corrected	0.423	0.035	11.936	7.637e-33	[0.354, 0.493]
Robust	0.423	0.038	11.168	5.870e-29	[0.349, 0.497]

So this is saying that if you exclude at 0 (no exclusions) or 0.005 our running variable isn't cleanly associated with "ever published" (at 0.0 coefficient is close to zero (and n.s.). at 0.05 it is negative (and n.s.). At 0.01 the effect is positive and robust (significant), so this justifies the donut

The flip in significance shows significant contamination at the cutoff (those just below *more* likely to be ever published than those just above!). So the justification for the donut is not just that it excludes problematic points, but the RDD is only possible against a background of linear stability (which would not exist without the donut).

Note also the negligible drop in N with donut 0.01. We're not losing too much data this way.

We can visualise the donut, and this in particular provides the justification (showing that our contamination of high $p(\text{ever_published})$ Notes below the cutoff is particularly high in the 0.39-0.40)



So this shows the operation for the $\text{inertiaDelta} = 0.01$ rule, see `scoring_rules.py` line 1521. Notes which are published and drop 0.01 or less keep their status <https://github.com/twitter/communitynotes/blob/>

The OLS serves as a sanity check, but now let's run the full fuzzy RDD

```
running variable (x) length 105375
binary outcome (y) length 105375
cutoff (c) is 0.4
Call: rdrobust
```

Fuzzy RD estimates using local polynomial regression.

```
Number of Observations:      105375
Polynomial Order Est. (p):    1
Polynomial Order Bias (q):    2
Kernel:                      Triangular
Bandwidth Selection:          mserd
Var-Cov Estimator:           NN
```

	Left	Right
Number of Observations	92694	12681
Number of Unique Obs.	92529	12672
Number of Effective Obs.	4994	5005
Bandwidth Estimation	0.065	0.065
Bandwidth Bias	0.177	0.177
rho (h/b)	0.366	0.366

Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	0.096	0.063	1.515	1.297e-01	[-0.028, 0.22]
Robust	-	-	1.475	1.402e-01	[-0.033, 0.235]

First-Stage Estimates.

Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	0.45	0.025	17.818	5.113e-71	[0.4, 0.499]
Robust	-	-	15.972	2.009e-57	[0.378, 0.484]

MSE-optimal h = 0.0648

Bias-corrected point estimate: 0.1007 95% CI [-0.0331, 0.2346]

Bandwidth sensitivity (bias-corrected point, robust 95% CI):

h = 0.0324 (0.5x) tau = +0.1521 95% CI [-0.1290, +0.4248] N_L/N_R = 1539/1967
h = 0.0648 (1.0x) tau = +0.0960 95% CI [-0.0331, +0.2346] N_L/N_R = 4994/5005
h = 0.1297 (2.0x) tau = +0.1011 95% CI [+0.0435, +0.1847] N_L/N_R = 16625/10207

This is positive but non-significant

6 Converting rate to baseline outcome rate

Common bandwidth h = 0.0648

Reduced form (ITT): +0.0432 robust CI [-0.0432, +0.2087]
First stage: +0.4499 robust CI [+0.3533, +0.5801]
Fuzzy LATE: +0.0960 robust CI [-0.0331, +0.2346]

Tie-out check: RF / FS = +0.0960 (should LATE at common h)

The RD robust method identifies a MSE-optimal bandwidth of $h = 0.065$, and provides an estimated treatment effect of $\downarrow \hat{\tau} = +0.096$. This reflects +9.6percentage point increase in the probability of writing a second note if an author's first note is published (conventional local-linear estimator; robust bias-corrected 95% CI [-3.3, +23.5], $p = 0.14$). Calonico et al., 2014

The rate of republishing for Notes which fall before the cutoff is ~69.2%. The treatment effect of 9.6pp, implies a 13.9% relative lift over the counterfactual baseline, however this difference is

7 Showing an illustrative RDD plot

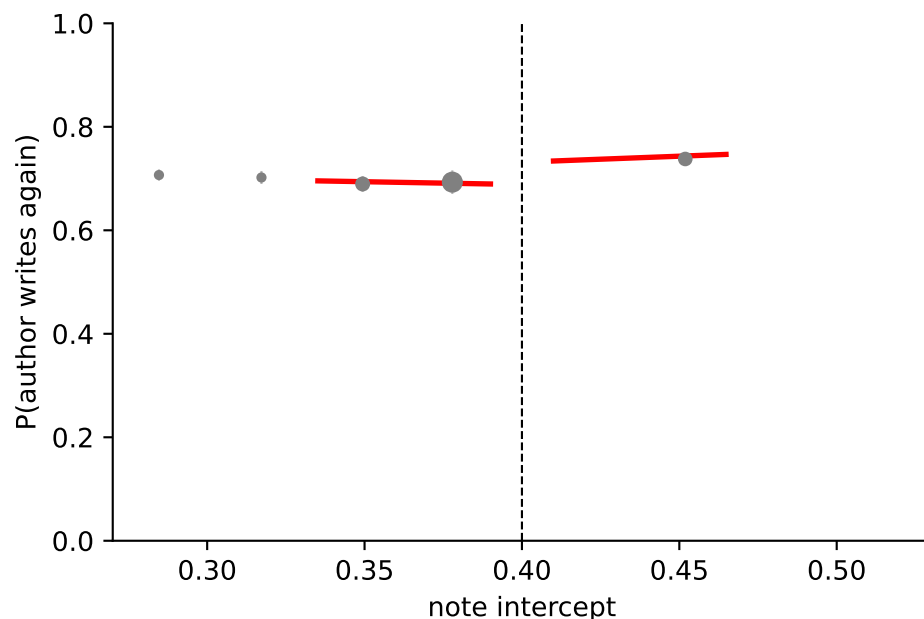
RDD: publication threshold at 0.4

Y-axis: P(author writes again)

X-axis: note intercept (current score)

Vertical line at x = 0.4

Now with the local-linear fits (over the `rdrobust` defined window), which more accurately represents what the model is doing



consistency check: gap at cutoff should conventional tau
 plot gap at $c = +0.0432$ vs conventional tau = +0.0960

8 Robustness check - density test

Next, manipulation test/density smoothness at cutoff. The density test is standard for RDD, to show that nothing weird is happening in *counts* at the cutoff.

This also, incidentally, shows the donut cut

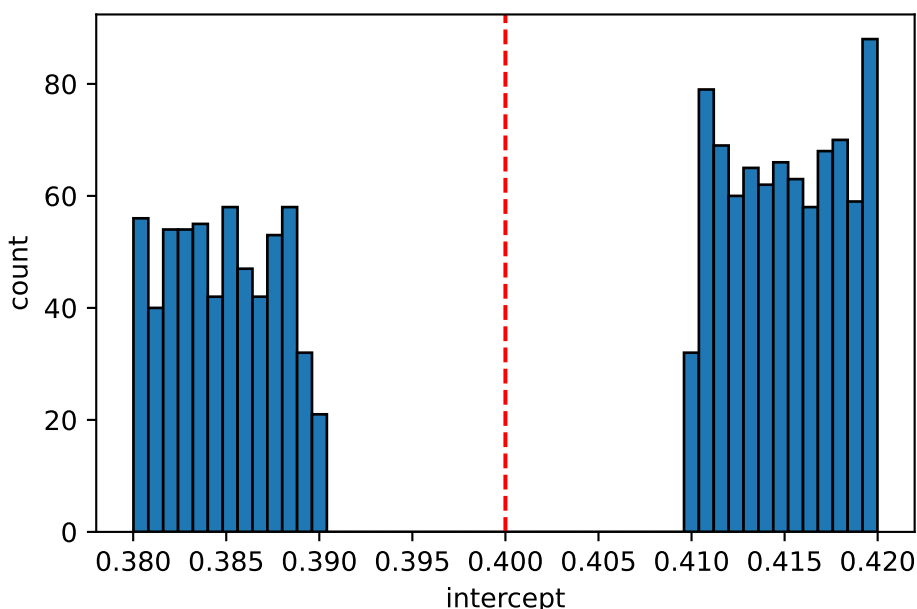
Inspecting density with a window of ± 0.02

```
t_asy      NaN
t_jk       0.31041
p_asy      NaN
p_jk       0.75625
dtype: float64
left       0.474619
right      0.530506
diff       0.055888
dtype: float64
left       0.050018
right      0.048857
dtype: float64
full       105375
left       92694
right      12681
eff_left   3207
eff_right  3506
dtype: int64
leftN      [612.0, 826.0, 1086.0, 1365.0, 1613.0, 1888.0, ...]
```

```

rightN      [839.0, 1112.0, 1397.0, 1694.0, 2008.0, 2308.0...
leftWindow   [0.02, 0.023335316347913374, 0.026670632695826...
rightWindow  [0.02, 0.0232063742081029, 0.02641274841620580...
pval         [2.768586098637157e-09, 8.849180397584868e-11,...
dtype: object
['X_max', 'X_min', 'binNW', 'bino', 'binoN', 'binoNStep', 'binoP', 'binoW', 'binoWStep', 'bino']

```



The density does progressively decrease up to the cutoff, but there are but no signs of sub-threshold notes being shifted to above threshold (decline not matched by leap post-cutoff)

9 Robustness check - Donut RDD

We can run the analysis with different size donuts. This isn't really necessary, since our donut size is defined from first principles (the 0.01 hysteresis on Notes getting unpublished), but let's have a look anyway).

donut	n_eff_L	n_eff_R	h	coef_conv	se_conv	p_robust	ci_robust_lo	ci_robust_hi
±0.000	2769	3068	0.035	-0.1527	0.3128	0.4796	-0.8781	0.4126
±0.005	2674	3129	0.040	-0.0159	0.1946	0.8416	-0.4486	0.3656
±0.010	4994	5005	0.065	0.0960	0.0633	0.1402	-0.0331	0.2346
±0.020	4979	4554	0.069	0.0299	0.0803	0.7468	-0.1473	0.2054

This shows that the effect never gets significant at any donut width (ie the non-effect is robust). You can see also that the donut = 0 row is aberrant (coefficient has negative sign), which is due to the contamination which justified a non-zero donut in the first place

10 Sensitivity check - RDD sensitivity with different cutoffs

Only a cutoff of 0.4 makes sense in terms of the CN publication mechanism. Therefore we expect other thresholds to show no effect (definitely below 0.4, cutoffs above 0.4 will include some published

notes above and below that cutoffs, so if there is a true effect may report an attenuated form of that. We can adjust for this by running the RDD on the data above 0.4 only for thresholds above 0.4)

```
running variable (x) length 105375
binary outcome (y) length 105375
cutoff (c) is 0.3
Call: rdrobust
```

Sharp RD estimates using local polynomial regression.

```
Number of Observations:      105375
Polynomial Order Est. (p):      1
Polynomial Order Bias (q):      2
Kernel:                        Triangular
Bandwidth Selection:            mserd
Var-Cov Estimator:             NN
```

	Left	Right
Number of Observations	82213	23162
Number of Unique Obs.	82066	23135
Number of Effective Obs.	13056	8189
Bandwidth Estimation	0.059	0.059
Bandwidth Bias	0.102	0.102
rho (h/b)	0.582	0.582

Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	0.007	0.014	0.525	5.993e-01	[-0.02, 0.034]
Robust	-	-	0.666	5.051e-01	[-0.021, 0.042]

MSE-optimal h = 0.0591

Bias-corrected point estimate: 0.0106 95% CI [-0.0206, 0.0418]

Bandwidth sensitivity (bias-corrected point, robust 95% CI):

```
h = 0.0296 (0.5x) tau = +0.0233 95% CI [-0.0138, +0.0715] N_L/N_R = 6122/4755
h = 0.0591 (1.0x) tau = +0.0072 95% CI [-0.0206, +0.0418] N_L/N_R = 13056/8189
h = 0.1182 (2.0x) tau = -0.0013 95% CI [-0.0201, +0.0252] N_L/N_R = 29667/11156
```

```
running variable (x) length 105375
binary outcome (y) length 105375
cutoff (c) is 0.35
Call: rdrobust
```

Sharp RD estimates using local polynomial regression.

```
Number of Observations:      105375
```

Polynomial Order Est. (p): 1
 Polynomial Order Bias (q): 2
 Kernel: Triangular
 Bandwidth Selection: mserd
 Var-Cov Estimator: NN

	Left	Right
Number of Observations	89487	15888
Number of Unique Obs.	89326	15875
Number of Effective Obs.	14706	5453
Bandwidth Estimation	0.085	0.085
Bandwidth Bias	0.13	0.13
rho (h/b)	0.656	0.656

Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	0.003	0.015	0.197	8.441e-01	[-0.026, 0.032]
Robust	-	-	0.29	7.720e-01	[-0.03, 0.04]

MSE-optimal h = 0.0855

Bias-corrected point estimate: 0.0051 95% CI [-0.0297, 0.0400]

Bandwidth sensitivity (bias-corrected point, robust 95% CI):

h = 0.0427 (0.5x) tau = +0.0173 95% CI [-0.0254, +0.0766] N_L/N_R = 5984/3207
 h = 0.0855 (1.0x) tau = +0.0029 95% CI [-0.0297, +0.0400] N_L/N_R = 14706/5453
 h = 0.1709 (2.0x) tau = +0.0032 95% CI [-0.0246, +0.0262] N_L/N_R = 37738/12828

running variable (x) length 12681

binary outcome (y) length 12681

cutoff (c) is 0.45

Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations: 12681
 Polynomial Order Est. (p): 1
 Polynomial Order Bias (q): 2
 Kernel: Triangular
 Bandwidth Selection: mserd
 Var-Cov Estimator: NN

	Left	Right
Number of Observations	3619	9062
Number of Unique Obs.	3617	9055
Number of Effective Obs.	1124	1156
Bandwidth Estimation	0.012	0.012

Bandwidth Bias	0.019	0.019
rho (h/b)	0.64	0.64

Method	Coef.	S.E.	z-stat	P> z	95% CI
-----	-----	-----	-----	-----	-----
Conventional	0.05	0.04	1.255	2.095e-01	[-0.028, 0.127]
Robust	-	-	1.225	2.206e-01	[-0.035, 0.15]

MSE-optimal h = 0.0122

Bias-corrected point estimate: 0.0577 95% CI [-0.0346, 0.1501]

Bandwidth sensitivity (bias-corrected point, robust 95% CI):

h = 0.0061 (0.5x)	tau = +0.0466	95% CI [-0.0888, +0.1749]	N_L/N_R = 599/612
h = 0.0122 (1.0x)	tau = +0.0496	95% CI [-0.0346, +0.1501]	N_L/N_R = 1124/1156
h = 0.0244 (2.0x)	tau = +0.0368	95% CI [-0.0259, +0.1042]	N_L/N_R = 2288/2245

Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations:	12681
Polynomial Order Est. (p):	1
Polynomial Order Bias (q):	2
Kernel:	Triangular
Bandwidth Selection:	mserd
Var-Cov Estimator:	NN

	Left	Right
-----	-----	-----
Number of Observations	3619	9062
Number of Unique Obs.	3617	9055
Number of Effective Obs.	1124	1156
Bandwidth Estimation	0.012	0.012
Bandwidth Bias	0.019	0.019
rho (h/b)	0.64	0.64

Method	Coef.	S.E.	z-stat	P> z	95% CI
-----	-----	-----	-----	-----	-----
Conventional	0.05	0.04	1.255	2.095e-01	[-0.028, 0.127]
Robust	-	-	1.225	2.206e-01	[-0.035, 0.15]

0.3 and 0.35 are non-significant and the effect estimates are close to zero

the 0.45 cutoff is significant, but the derived bandwidth is too small to be robust, suggesting degenerate model fitting

11 Sensitivity check - data for different time windows

We skip this, since there is no positive result to interrogate